Week 12 – Hypothesis Generation, Thinking Habits and Metacognition (Lean Notes)

No transcript was available for this week, so these notes are built from the Week 12 assignment and its answer key only. They cover everything the assignment tests; lecture detail beyond it may be missing. ⭐ = tested in the assignment (every card is).

1. Concept cards

1. Course-based vs research-based education ⭐

- **Course-based:** deals with an **existing body of knowledge**, follows a **prescribed syllabus and curriculum**.
- **PhD research:** expected to **create new knowledge**, does **not** follow a common fixed syllabus.
- The two are **not evaluated in the same way** (exams test known material; research is judged on originality and quality).

2. A PhD is self-directed ⭐

- A scholar cannot expect the guide to give a **complete step-by-step procedure**. PhD research is **largely self-directed** and the scholar must learn to **decide how it should progress**.
- No fixed syllabus does **not** mean no guidance: the guide **supports when ambiguity or difficulty arises**.

3. What makes a scientific hypothesis ⭐

- A hypothesis is an **educated prediction** based on existing knowledge and experience, **not a statement already established as true**.
- It must be **falsifiable**: there must be a possibility of **being disproved**.
- You may **not** protect it from contradicting observations with **untestable explanations**, and its truth is **not established before** experiments (otherwise there is nothing to test).

4. Falsifiability: the swan example ⭐

- Hypothesis: "**All swans are white.**" **One black swan disproves it**, but no number of white swans can prove it for ever.
- So you can examine it **without observing every swan**. The criterion stresses the **possibility of disproof**, not permanent truth.

5. From a broad topic to a testable hypothesis ⭐

- Recipe: (1) **ask relevant questions** about the topic and **why it matters**; (2) **refine the questions through repeated cycles** of literature exploration and observation; (3) then reach a testable hypothesis.
- Wrong: starting an **unrestricted literature survey as the only route**, or writing the **final hypothesis immediately** from the broad topic.

6. Anomalies give hypotheses ⭐

- An **anomaly** is when **observations do not match a commonly assumed explanation**. In the **obesity and eating-habit example**, observed influences such as the **amount of food offered** were a lead for further investigation.
- A **missing link or contradiction** in the literature gives scope for research and helps generate a hypothesis. The hypothesis came from **literature and observations**, not directly from the broad topic.

7. Revised Bloom's Taxonomy ⭐

Order	Levels
Lower-order	Remember, Understand, Apply
Higher-order	Analyze, Evaluate, **Create** (top)

- Create sits at the **top** because it needs the **most** thinking. The taxonomy does **not** say every person finds all six levels equally hard.

8. Bloom's levels in PhD work ⭐

- **Analyze:** collecting data and examining unfamiliar patterns (analysis is an important **starting level** in PhD research).
- **Evaluate:** considering **alternative hypotheses** and trying to validate them.
- **Create:** developing a **new research idea**.
- **All six levels** can be used while solving a research problem; **memorising** is a lower-order activity and is not the main focus of a PhD.

9. Creativity and functional fixedness ⭐

- **Creativity can be developed through practice**; it is not a fixed ability at birth.
- **Functional fixedness** = seeing an object **only through its conventional use** (a brick only as a building block). It **hinders lateral thinking** and new approaches.
- Knowing the **whole existing body of knowledge does not guarantee** creating something new.

2. Top traps (most likely first)

1. **Falsifiable means "could be shown false"**, not "proved true" and not "protected by untestable excuses".
2. **Q9: all three correct options are A, B and D** (A, "all six levels may be used", is easy to miss).
3. **Create is the top** Bloom level; Remember, Understand, Apply are lower-order.
4. **Memorising is not the focus of a PhD.**
5. **Functional fixedness hurts**; it does not help with novel challenges.
6. **No fixed syllabus does not mean no guide.**
7. Do not jump from a broad topic to a final hypothesis; **refine questions by repeated reading and observing**.
8. "Identical" or absolute wording ("exactly the same way", "guarantees", "every swan") makes an option wrong.

3. Assignment questions with solutions

Q1. Which statements correctly distinguish course-based education from research-based education? (select all that apply) *(Q1)* - A. Course-based education primarily deals with an existing body of knowledge, whereas PhD research is expected to create new knowledge B. A PhD programme follows a common prescribed syllabus and curriculum in the same way as a course-based degree C. Course-based education generally has a prescribed syllabus and curriculum, whereas research-based education does not simply follow a fixed syllabus D. Knowledge acquired in course-based education and new knowledge generated through PhD research are evaluated in exactly the same way

Solution (card 1): B and D claim the two are the same. **Answer: A and C.**

Q2. A research scholar beginning a PhD expects the guide to provide a complete step-by-step procedure for the entire programme. Which responses are consistent? (select all that apply) *(Q2)* - A. The scholar should recognize that PhD research involves substantially self-directed work B. The absence of a fixed syllabus means that the scholar does not need guidance from the research guide C. The research guide can provide guidance and support when

ambiguity or difficulty arises D. The scholar is expected to develop the ability to decide how the research should progress

Solution (card 2): B is wrong; guidance still exists. **Answer: A, C and D.**

Q3. A researcher proposes a hypothesis based on existing knowledge and experience. Which conditions are relevant to treating it as a scientific hypothesis? (select all that apply) *(Q3)* - A. It should be an educated prediction rather than a statement already established as true B. It should allow a possibility of being disproved C. It should be protected from contradictory observations by adding explanations that cannot be tested D. Its truth must be established before experiments or data collection begin

Solution (card 3): C makes it unfalsifiable and D leaves nothing to test. **Answer: A and B.**

Q4. Consider the hypothesis "All swans are white." Based on the falsifiability example, which statements are correct? (select all that apply) *(Q4)* - A. Finding a black swan would provide an observation capable of disproving the hypothesis B. It is necessary to observe every swan in the world before the hypothesis can be scientifically examined C. The possibility of disproving the hypothesis is central to the criterion discussed D. The criterion emphasizes the possibility of disproof rather than establishing the hypothesis as permanently true

Solution (card 4): B is wrong. **Answer: A, C and D.**

Q5. A student starts a PhD with a broad topic. Which actions are appropriate before arriving at a testable hypothesis? (select all that apply) *(Q5)* - A. Begin by formulating relevant questions about the topic and considering why the topic is important B. Start an unrestricted literature survey immediately and use it as the only route for identifying the research problem C. Use literature exploration and observations to refine the initial questions through repeated cycles D. Formulate the final testable hypothesis immediately from the broad topic without further exploration

Solution (card 5): A then C; B and D are premature or unfocused. **Answer: A and C.**

Q6. In the obesity and eating-habit example, observations from the literature and study were used to identify an anomaly and develop a hypothesis. Which statements are supported? (select all that apply) *(Q6)* - A. An anomaly can arise when observations do not match a commonly assumed explanation B. The observed influence of factors such as the amount of food offered can motivate further investigation C. The hypothesis was formulated directly from the broad topic without literature survey or observations D. Identifying a missing link or contradiction can provide scope for further research and help generate a hypothesis

Solution (card 6): C ignores the literature and observations. **Answer: A, B and D.**

Q7. According to the Revised Bloom's Taxonomy, which statements are correct? (select all that apply) *(Q7)* - A. Remember, Understand and Apply are described as lower-order thinking levels B. Analyze, Evaluate and Create are described as higher-order thinking levels C. Create is placed below Remember because generating a new idea generally requires less cognitive effort D. The taxonomy states that every individual experiences exactly the same difficulty across all six levels

Solution (card 7). **Answer: A and B.**

Q8. A PhD scholar collects experimental data, examines unfamiliar patterns, considers alternative hypotheses, and develops a new research idea. Which Bloom's levels are most directly represented? (select all that apply) *(Q8)* - A. Analyze B. Evaluate C. Remember D. Create

Solution (card 8): examining patterns is analysis, weighing alternatives is evaluation, a new idea is creation. **Answer: A, B and D.**

Q9. Which statements reflect the role of higher-order thinking in PhD research? (select all that apply) *(Q9)* - A. All six levels of thinking may be used during research problem solving B. The lecture identifies analysis as an important starting level in PhD research because researchers collect and analyze data C. The main intellectual focus of a PhD is

memorizing information from the existing body of knowledge D. Evaluation can involve considering alternate hypotheses and attempting to validate them

Solution (card 8): C is wrong. **Answer: A, B and D** (picking only two of these earns partial credit).

Q10. Which statements are consistent with the discussion of creativity and functional fixedness? (select all that apply) *(Q10)* - A. Creative thinking can be developed or improved through practice rather than being limited to an ability fixed at birth B. Functional fixedness can cause a person to perceive an object only through its conventional purpose and can hinder lateral thinking C. Complete knowledge of the existing body of knowledge guarantees the creation of something new D. Functional fixedness helps researchers generate new approaches when facing unexpected or novel challenges

Solution (card 9): C says “guarantees” and D reverses the effect. **Answer: A and B.**